

DBLP Explorer

A Graph-Theoretic Analysis of the Computer Science
Co-Authorship Network, Derived from Real dblp Bibliographic Data

Course Project Report

Course: Graph Theory (M2)
Instructor: Dr. Hussein Hazimeh
Submitted by: Abbass Zahreddine
September 2026

377,243
graph nodes (authors)

2,695,430
weighted edges (co-authorships)

10,711
communities detected

334,137
papers analyzed

Abstract

This project constructs and analyzes the co-authorship network of a real subset of the dblp computer science bibliography using classical graph-theoretic measures. Authors are modeled as nodes and joint publications as weighted, undirected edges, yielding a real graph of 377,243 nodes and 2,695,430 edges built from 334,137 papers across 38 well-known artificial intelligence, database, and data-mining venues. Two centrality measures from the literature - PageRank (Brin and Page, 1998) and betweenness centrality (Freeman, 1977), computed via Brandes' algorithm (Brandes, 2001) - are applied to identify influential and bridging authors respectively. Louvain community detection (Blondel et al., 2008) partitions the network into research communities, and shortest-path queries demonstrate the network's small-world structure. A central finding of this project is empirical rather than theoretical: exact betweenness centrality, while well defined and implementable, proved computationally infeasible on this graph within a practical time budget, motivating a measured, evidence-based transition to sampled betweenness. The resulting rankings were validated by inspection against well-known researchers in the field, and the full pipeline was built into an interactive web application allowing exploration of the network, its communities, and shortest paths between authors.

1. Introduction and Objectives

Bibliographic co-authorship networks are a canonical application domain in graph theory and social network analysis: nodes represent authors, and an edge between two authors indicates they have co-written at least one publication together. Such networks have been studied extensively since Newman's foundational analysis of scientific collaboration networks (Newman, 2001), which showed that these networks exhibit small-world properties, a heavy-tailed degree distribution, and identifiable community structure.

The objective of this project was to construct such a network from real, unmodified bibliographic data - not a synthetic or toy graph - and to apply the core measures covered in this course to it: degree, PageRank, betweenness centrality, and community detection, followed by shortest-path analysis between individual authors. A secondary objective, which turned out to be at least as instructive as the primary one, was to observe where the textbook algorithms meet the real constraints of a graph at genuine scale - in particular, the gap between betweenness centrality's polynomial-time exact algorithm and what is actually tractable on a graph with hundreds of thousands of nodes.

2. Graph Construction

The dblp computer science bibliography is distributed as a single, monolithic XML file listing every indexed publication. The September 2026 release, 1,099,942,087 bytes, was streamed and parsed in full, yielding 12,901,616 individual bibliographic records. A domain-restricted subset was then selected for analysis: 38 venues recognized in the artificial intelligence, database, and data-mining sub-fields (including AAAI, NeurIPS, ICML, ACL, CVPR, KDD, SIGMOD, VLDB, and SIGIR), restricted to publications from the year 2000 onward. This yielded 334,137 papers - a scope chosen specifically because it forms a coherent, recognizable research community, which matters directly for validating centrality results by inspection later in this report.

The co-authorship graph itself is derived formally as follows. Let P be the set of papers in scope and $A(p)$ the ordered set of authors of paper p . The weighted, undirected co-authorship graph $G = (V, E, w)$ is defined by:

$$\begin{aligned} V &= \{ a : a \in A(p) \text{ for some } p \in P \} \\ E &= \{(u, v) : \exists p \in P \text{ such that } u, v \in A(p), u \neq v\} \\ w(u, v) &= |\{p \in P : u, v \in A(p)\}| \end{aligned}$$

That is, an edge's weight is the number of papers the two endpoint authors co-authored together. This construction was implemented as a materialized view inside the underlying relational database, computed directly from the authorship join table, and yields the real graph statistics reported throughout this document: 377,243 nodes and 2,695,430 weighted edges.

2.1 Node identity: a real data-quality issue in graph construction

A subtlety specific to constructing a graph from real bibliographic data, rather than a pre-cleaned dataset, is author disambiguation: two records referring to the same person must map to the same node, and two different people who happen to share a name must not be merged into one. dblp addresses the latter by appending a numeric disambiguation suffix to homonymous authors' names (e.g. "Wei Wang 0001"); 32,949 distinct base names carrying such a suffix were found in the source data, confirming this is a real and non-trivial concern at this scale, not an edge case.

A second, subtler issue was found during construction: dblp records an author's persistent identifier (ORCID) on some of that author's publications but not consistently on all of them. Because node identity was initially keyed on this identifier when present, the same real person could be - and in two confirmed real cases was - represented as more than one distinct node. One of the two most prolific authors in the resulting graph was found to be split across three separate node identities for exactly this reason. This is reported here as a genuine limitation of the graph as constructed: a complete fix would require resolving every author alias against dblp's own person-record index, which was outside this project's scope, but its effect on the specific rankings reported below is disclosed explicitly wherever it applies.

3. Centrality Analysis

Two centrality measures were computed on the co-authorship graph, chosen because they answer different structural questions: PageRank identifies *influential* nodes, well-connected to other well-connected nodes, while betweenness centrality identifies *bridging* nodes that lie on many shortest paths between other pairs of nodes. A node can score highly on one measure and low on the other.

3.1 PageRank

PageRank (Brin and Page, 1998) models a random walker who follows edges with probability proportional to edge weight and, with probability $1 - d$, jumps to a uniformly random node. The stationary distribution of this walk defines each node's score:

$$PR(v) = (1 - d)/N + d \cdot \sum_{u \in N(v)} [w(u, v) / \sum_{x \in N(u)} w(u, x)] \cdot PR(u)$$

with damping factor $d = 0.85$, computed by power iteration until convergence. On the real graph, PageRank converged in 9.3 seconds combined with community detection (below) - well within the polynomial time bound expected of an $O(E)$ per-iteration algorithm even at this scale.

3.2 Betweenness centrality

Betweenness centrality (Freeman, 1977) measures how often a node lies on the shortest path between two other nodes:

$$C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$$

where σ_{st} is the number of shortest paths from s to t and $\sigma_{st}(v)$ is how many of them pass through v . Brandes' algorithm (Brandes, 2001) computes this exactly for all nodes in $O(VE)$ time on an unweighted graph by accumulating dependency scores from a single-source shortest-path search rooted at every node in turn.

3.3 A real computational limit, and its resolution

$O(VE)$ is polynomial, and therefore "efficient" in the formal complexity-theoretic sense. It is nonetheless not tractable in practice on this graph: with $V = 377,243$ and $E = 2,695,430$, an exact run requires on the order of 10^{12} elementary operations. This was confirmed empirically, not assumed: an exact computation, and even a depth-limited variant restricted to shortest paths of at most two hops, both failed to complete within twenty real minutes on this graph.

The standard remedy in the literature is sampled (approximate) betweenness: rather than rooting a shortest-path search at every node, root it at only k randomly chosen source nodes and scale the resulting partial sums by N/k , giving an unbiased estimator of the true betweenness score. Applying this required choosing k . Rather than adopt a default value uncritically, per-source cost was measured directly on the real graph: a consistent 8.65 seconds per source, independent of k in the range tested. A commonly cited default of $k = 1,000$ would therefore require approximately 144 minutes - acceptable for a one-off analysis, but incompatible with a repeatable pipeline. $k = 150$ was selected instead as the largest sample size fitting comfortably inside a practical time budget, and the resulting betweenness rankings, reported in Section 6, were still found to agree closely with intuition about who the network's real bridging figures are.

This finding - that an algorithm's textbook complexity class does not, by itself, guarantee practical tractability at real-world scale, and that the standard sampling remedy requires its own empirical calibration rather than a default parameter - is presented here as one of this project's more instructive results.

3.4 Community detection

The Louvain method (Blondel et al., 2008) was applied to the same weighted graph to detect communities by greedy modularity maximization. It identified 10,711 communities, ranging from a small number of very large communities (the largest containing 33,403 authors) down to many small, tightly-knit groups - consistent with the community-size distributions reported for real collaboration networks in the literature. Community detection completed as part of the same 9.3-second combined run as PageRank, in contrast to the far more expensive betweenness step, illustrating a second practical complexity gap between different graph algorithms applied to the same real graph.

4. Shortest Paths and Network Distance

An unweighted breadth-first search, capped at a maximum of six hops, was implemented to answer "Erdos number"-style queries between any two authors in the network: the shortest chain of co-authorship relationships connecting them, together with a real joint publication justifying each step of the path.

As a concrete illustration, the two authors Philip S. Yu and Yoshua Bengio - both highly ranked by PageRank but not direct co-authors - were found to be connected by a path of length three:

Philip S. Yu	→	Zhiping Xiao	→	Jian Tang	→	Yoshua Bengio
--------------	---	--------------	---	-----------	---	---------------

Each edge in this path corresponds to a real, specific joint publication retrieved from the dataset.

A single path of this kind is illustrative rather than statistically conclusive, but it is consistent with the small-world property widely reported for real collaboration networks (Newman, 2001): most author pairs, even those working in visibly different sub-fields, are connected through only a small number of intermediate collaborators.

5. Network Structure: Degree Distribution and Productivity

The distribution of authors per paper and papers per author was computed directly from the source data. The mean number of authors per paper is 3.46 (median 3), while the number of papers per author is heavily right-skewed: the mean is 5.73, but the median is only 2, and 54% of all authors in the network have exactly one paper in scope. This long-tailed distribution - a small number of extremely prolific authors alongside a very large number with minimal output - is the hallmark of the scale-free, power-law degree distributions reported for real collaboration networks (Newman, 2001), and was confirmed here directly from real authorship counts rather than assumed from prior literature.

6. Results and Validation

The most direct validation available for centrality rankings on a real network of this kind is inspection: do the top-ranked nodes correspond to individuals genuinely recognized as prominent or well-connected within the chosen field? Both rankings below were computed on the real graph described in Section 2.

6.1 Top authors by PageRank

Rank	Author	Notes
1	Philip S. Yu	See Section 2.1 regarding a three-way node-identity split affecting this author
2	Jiawei Han	
3	Luc Van Gool	
4	Radu Timofte	Affected by the same node-identity issue (Section 2.1)
5	Yoshua Bengio	

6.2 Top authors by betweenness centrality

Rank	Author
1	Philip S. Yu
2	Jiawei Han
3	Charu C. Aggarwal
4	Luc Van Gool
5	Jiliang Tang

Every individual appearing in both tables is a genuinely prominent, independently verifiable researcher in the artificial intelligence, data mining, and computer vision communities that dominate the chosen venue set - the qualitative validation this type of ranking calls for. The presence of the same node-identity limitation in both tables (Section 2.1) is reported transparently rather than corrected after the fact, since doing so would require re-running the underlying algorithms against a corrected graph, which was not performed for this report.

6.3 Performance summary

Stage	Measured Time
PageRank + Louvain community detection (combined)	9.3 seconds
Betweenness centrality, exact (did not complete)	> 20 minutes (aborted)
Betweenness centrality, sampled (k = 150)	21.2 minutes
Total centrality pipeline	21.4 minutes

7. Implementation Overview

While the focus of this report is the graph-theoretic methodology and its results, the pipeline above was implemented as a complete, working system rather than a one-off script, so that the network could be explored interactively rather than only summarized statically. The co-authorship graph is stored inside a PostgreSQL database as a materialized view derived from the raw authorship data; PageRank and Louvain community detection were computed using the *igraph* library; betweenness centrality used the sampled-source approach described in Section 3.3; and the results were exposed through a web interface with a force-directed network visualization (built on *Sigma.js* and the *ForceAtlas2* layout algorithm), sortable centrality leaderboards, and an interactive shortest-path and ego-network explorer implementing the queries described in Section 4.

8. Discussion and Limitations

- **Node identity.** As discussed in Section 2.1, a small number of real authors are split across more than one node due to inconsistent identifier tagging in the source data. This affects the magnitude, though not the overall qualitative conclusion, of the centrality results in Section 6 for the specific individuals concerned.
- **Sampled betweenness is an approximation.** The betweenness scores in Section 6.2 are statistical estimates, not exact values; a different random seed or a larger sample size would produce numerically different, though expected to be qualitatively similar, rankings.
- **Scope restriction.** The graph analyzed here covers a curated 38-venue, post-2000 subset of *dblp* rather than the full bibliography, chosen deliberately for thematic coherence; results should be read as characterizing this sub-community rather than computer science as a whole.

9. Conclusion

This project applied PageRank, betweenness centrality, Louvain community detection, and shortest-path analysis to a real, non-trivial co-authorship graph of 377,243 authors and 2,695,430 weighted edges, constructed directly from the *dblp* bibliography rather than a synthetic dataset. The resulting rankings were qualitatively validated against known figures in the chosen research community, and the network's shortest-path structure was demonstrated to be consistent with the small-world property reported in the literature. Beyond applying the algorithms themselves, the project's most substantive finding was empirical: betweenness centrality's polynomial-time complexity does not, in practice, make it tractable on a graph at this scale, and the standard sampling remedy requires its own measured calibration rather than an assumed default - a concrete illustration of the gap between a graph algorithm's asymptotic complexity class and its real-world tractability.

References

- Brin, S. and Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1-7), 107-117.
- Freeman, L. C. (1977). A set of measures of centrality based on betweenness. *Sociometry*, 40(1), 35-41.

Brandes, U. (2001). A faster algorithm for betweenness centrality. *Journal of Mathematical Sociology*, 25(2), 163-177.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008.

Newman, M. E. J. (2001). The structure of scientific collaboration networks. *Proceedings of the National Academy of Sciences*, 98(2), 404-409.